Conjecture 00000004307: incompatible necessary rank formulas

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1 The literal conjunction

The English source asserts:

> The minimal rank of an Euler system for the higher K-group is $2k-1$, and when k is coprime to $p-1$ the rank is exactly 1, the formula being complete.

The Chinese source gives the same two numerical clauses. We refute their literal conjunction for the *same rank*. A reading in which the conditional clause overrides the first formula, defining a corrected piecewise function, is a different assertion and is not refuted here. The complete original source accompanies this submission.

2 A legal witness and the contradiction

Choose $k = 3$, $p = 5$. Then $k > 0$, p is an odd prime, $k \le p-2$, and $\gcd(k, p-1) = \gcd(3, 4) = 1$. If r denotes the one rank referred to by the two clauses, they require simultaneously

$$r = 2 \cdot 3 - 1 = 5, \qquad r = 1,$$

which is impossible. Even imposing the additional restriction $1 \le k \le p-2$ on both formulas, or fixing the single odd prime $p = 5$, leaves this contradiction. That restriction only weakens the required universal clauses; it is not an added hypothesis needed from the source.

The proof does not assign a value to an Euler-system rank. Rather, it proves that *every* candidate rank function fails at least one of the two necessary clauses. Consequently no actual mathematical construction could satisfy their conjunction, regardless of its true rank. No Euler system or higher K-group is replaced with an artificial object, and no theorem computing their actual ranks is claimed.

3 Formal statement and verification

Lean proves the pointwise obstruction for every value in every characteristic-zero semiring, then quantifies over all functions $\rho : \mathbb{N} \times \mathbb{N} \to R$. It proves both the restricted and unrestricted universal-formula negations and the fixed-$p = 5$ negation. Explicit specializations cover natural, integer and cardinal-valued candidate ranks, including the ordinary cardinal notion of rank. The named predicates are only the two numerical necessary conditions; they are not definitions of Euler systems.

Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

are pinned. The package includes a new root build, transitive axiom audits of all nine theorems and full official `leanchecker --fresh Main` replay. No `sorry`, `native_decide` or additional axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed the source, proof, report and rendered PDF. This is AI review, not human certification. Organizer acceptance is pending.